

Detecting Unsoundness in Neural Network Verifiers via Concrete–Abstract Consistency

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Verifiers can be silently unsound

Verifiers certify networks.

But who certifies the verifier?

- A verifier **guarantees** no violation in a region — yet it is complex software, so it can be silently unsound: certify "safe" while a real counterexample exists.

Our goal: not to build a new verifier, but to test whether NN verifiers are sound.

This is not hypothetical — VNN-COMP 2023 — a tool reported 100% of a ViT benchmark verified, later found unsound; VNN-COMP 2025 — still records unsound results.

Our idea: concrete–abstract consistency

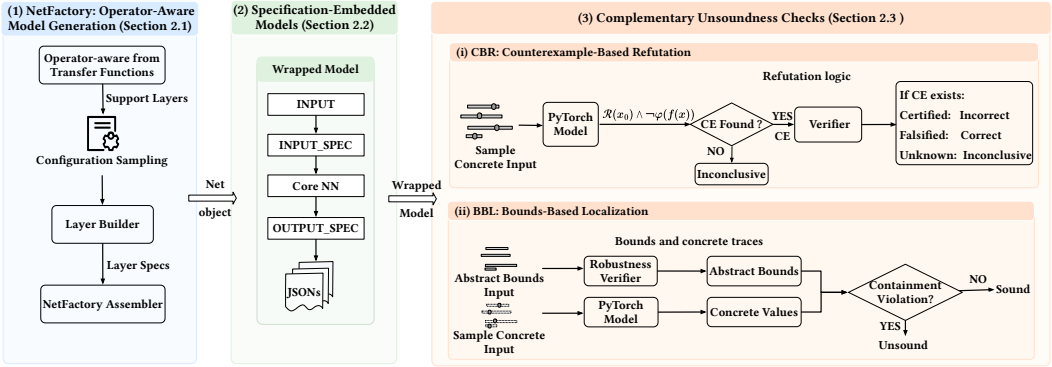
Gap. Existing checks (e.g. SoundnessBench) tell you a verdict is wrong — not where or why inside the verifier.

Contribution

We detect false certificates and localize the unsound layer / operator — via **concrete–abstract consistency**.

Abstract bounds must over-approximate concrete execution. We check this at two levels — **outputs** (CBR) and **per-layer bounds** (BBL).

Our framework: unsoundness detection framework



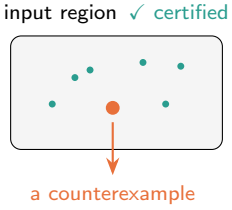
We verify the verifier: one spec-embedded instance feeds the verifier; **CBR** checks its **verdict**, **BBL** checks its **bounds**.

Two checks: CBR refutes, BBL localizes

CBR — *is the certificate false?*

certified $\wedge\ \exists x$:

$$\neg\varphi(f(x)) \ \Rightarrow\ \text{false certificate}$$

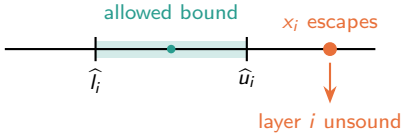


Evidence: a concrete witness.

BBL — *where did the abstraction break?*

$x_i \notin [\widehat{l}_i, \widehat{u}_i]$ at layer i $\Rightarrow$

localize the violated layer/operator



Evidence: the layer / operator location.

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Results

450 instances = 3 domains × 5 bug types × 30 networks

domains: interval · HybridZ · dual bug types: tighten · swap · zero · scale · noise

72%
detection

100%
top-1 localization
(among detected BBL violations)

1–2 ms
overhead / network
(at default budget 20)

CBR 15% output-visible false certs · **BBL 72%** internal bound violations.

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Takeaway: certify the certifier

CBR tells us the certificate is wrong;
BBL tells us where the verifier goes wrong.

- Test the **verifier**, not only the network.
- Concrete–abstract consistency **detects** false certificates and **localizes** violated / unsound operators.

Open Source: `github.com/SVF-tools/ACT`

Thank you! Questions?

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